R&D | RetailPulse

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SQL Analysis Specification

> **In simple words**
>
> This is the to-do list for SQL. Each question has a business purpose, a hypothesis to test, the SQL plan, and the insight to look for.

1. SQL rules for this project (PostgreSQL)

- One numbered `.sql` file per step. Add a comment header: purpose, input, output.
- Build in layers using views: clean data, then holiday flags, then baseline, then error table, then analysis.
- Use CTEs for readability. Use window functions for the baseline and for ranking.
- Use `EXTRACT(MONTH FROM date)` for months. Use `NULLIF` to avoid divide by zero.
- Check every result: totals must match across queries.

> **Four traps to avoid**
>
> **1. Holiday join:** use the store-date flag table (Doc 03).
> **2. Missing dates:** check that the earlier date is exactly 7 days before, do not trust row order.
> **3. Store not open yet:** exclude rows before a store's first sale date.
> **4. Zero sales:** use WAPE, not MAPE.

2. Build steps before analysis

File	What it builds	Check
01_create_tables.sql	Tables for the 5 CSVs	Columns and types match Doc 03
02_load_data.sql	Load CSVs	Row counts match Kaggle
03_data_checks.sql	NULLs, duplicates, missing dates, store start dates	Issues listed and handled
04_holiday_flag_table.sql	One row per store and date with is_holiday	Row count = stores x dates, no duplicates
05_baseline_forecast.sql	Baseline A and B using LAG / window functions	Earlier date is exactly 7 days before
06_error_table.sql	Fact table: actual, forecast, abs error, excess, shortfall, flags	Total sales equals train total after exclusions

3. Analysis questions

Each question has a **hypothesis** (our starting guess). We write it before running SQL, then test it. A hypothesis can be wrong. That is also an insight.

Q1. How healthy is the baseline overall?	
Business question	Overall, how accurate is the baseline, and is it getting better or worse over the years?
Hypothesis	Overall WAPE is moderate, and it does not change much between years.
Method	SUM abs error / SUM actual overall and by year. Add Bias %, excess and shortfall units.
Output	`overall_health.csv` : year, WAPE, Accuracy, Bias %, excess units, shortfall units.
Insight to look for	Is the problem everywhere or only in some places? This sets up the next questions.

Q2. Which families are hardest to predict?	
Business question	Which families have the highest WAPE, ignoring tiny-volume families?
Hypothesis	Low-volume and special families (for example books, baby care) have the highest WAPE.
Method	WAPE by family. Add total sales. Apply a minimum volume rule. RANK by WAPE.
Output	`family_error_summary.csv` : family, total sales, WAPE, Accuracy, Bias %, high_error_flag.
Insight to look for	Are high-error families big or small in sales? Big ones matter more.

Q3. Which families cause most of the error? (Pareto)	
Business question	Which few families make up 80% of total error?
Hypothesis	A small group of high-volume families causes most of the error units.
Method	Abs error by family, share of total, `SUM() OVER (ORDER BY ... )` for cumulative share, mark the 80% cut.
Output	`family_pareto.csv` : family, abs error, error share %, cumulative share %.
Insight to look for	The shortlist to fix first. This is usually your strongest recommendation.

Q4. Are we too high or too low? (Direction)	
Business question	For each family, do we mostly over-forecast (excess) or under-forecast (shortfall)?
Hypothesis	Some families lean to excess and others to shortfall. The pattern is not the same for all.
Method	Excess units and shortfall units by family. Bias % by family. Label as Over, Under or Balanced.
Output	`family_direction.csv` : family, excess units, shortfall units, Bias %, direction label.
Insight to look for	Different actions: cut orders where excess is high, add buffer where shortfall is high.

Q5. Which regions are hardest?	
Business question	Which states, cities, store types and clusters have the highest WAPE and Bias?
Hypothesis	Smaller cities and some store types have higher WAPE because volumes are lower.
Method	Join error table to stores. WAPE, Bias % and total sales by state, city, type, cluster. RANK.
Output	`region_performance.csv`: state, city, type, cluster, sales, WAPE, Bias %, rank.
Insight to look for	Is error linked to size (low volume) or to location? Compare WAPE with sales.

Q6. Which months and seasons are hardest?	
Business question	How does error change by month, quarter and named periods (Christmas, Carnival, school start, Mother's Day)?
Hypothesis	December has the highest error because of Christmas demand.
Method	WAPE and Bias % by month and quarter. CASE WHEN to label named periods. Compare each with the overall WAPE.
Output	`season_error.csv`: month, quarter, period label, WAPE, Bias %, sales.
Insight to look for	Which period needs extra planning, and does it lean to excess or shortfall?

Q7. What is the effect of holidays and events?	
Business question	Is error higher on holidays, and one week after a holiday?
Hypothesis	Error is higher on holidays. Exactly one week after a holiday, the forecast is too high (excess), because the baseline copies the holiday spike.
Method	Use the holiday flag table. WAPE and Bias % for holiday vs regular days. Add a flag for days whose baseline day (7 days earlier) was a holiday.
Output	`holiday_impact.csv`: group (holiday, week-after-holiday, regular), WAPE, Bias %, sales.
Insight to look for	A clear rule: the baseline breaks in two places, on the holiday and one week later.

Q8. What is the effect of promotions?	
Business question	Does error rise when items are on promotion?
Hypothesis	Promotion days have higher WAPE and lean to shortfall, because sales jump above last week.
Method	Group rows by onpromotion = 0 and onpromotion above 0. WAPE, Bias % by family group.
Output	`promotion_impact.csv`: promo group, WAPE, Bias %, shortfall units.
Insight to look for	Whether promotions should feed the forecast as an input.

Q9. Is there a payday effect?	
Business question	Is error higher around the 15th and the last day of the month?
Hypothesis	Paydays raise sales and the baseline under-forecasts them.
Method	Flag the 15th and last day (and one day after). WAPE and Bias % for flagged vs other days.
Output	`payday_impact.csv`: payday group, WAPE, Bias %, sales.
Insight to look for	A simple calendar rule that the planners can use.

Q10. Which weekdays are hardest?	
Business question	Does error change by day of the week?
Hypothesis	Weekend days have more error than weekdays.
Method	`EXTRACT(DOW FROM date)`. WAPE and Bias % by weekday and by family group.
Output	`weekday_error.csv`: weekday, WAPE, Bias %, sales.
Insight to look for	Whether the stock plan needs a weekday adjustment.

Q11. Which baseline is better for which family?	
Business question	Is Baseline B (4-week average) better than Baseline A (last week) for some families?
Hypothesis	B is better for smooth families. A is better for families that move fast.
Method	WAPE for A and B by family. Compute the difference. Label the better one.
Output	`baseline_compare.csv`: family, WAPE A, WAPE B, better baseline.
Insight to look for	A practical tip: use B for volatile families. This is a simple comparison, not model building.

Q12. Is the error getting better or worse?	
Business question	Does WAPE trend up or down over time for the top families?
Hypothesis	Error is stable over time, except in event months.
Method	WAPE by month and year for the top 5 families. Use window functions for a 3-month rolling WAPE.
Output	`error_trend.csv`: year-month, family, WAPE, rolling WAPE.
Insight to look for	A stable problem needs a process fix. A growing problem needs urgent action.

4. SQL skills shown by this project

Technique	Used in
JOINs (inner, left)	Q5, Q7 and the holiday flag table
CTEs	Baseline, error table, Q3, Q11
Window functions: LAG, AVG OVER, SUM OVER, RANK	Baseline, Q2, Q3, Q5, Q12
CASE WHEN	Direction labels, periods, flags
Date functions: EXTRACT, DATE_TRUNC	Q6, Q9, Q10, Q12
Views and clear script order	All steps
Data quality checks	Step 03 and the QA list

5. Output files for the report and checks

These CSV files go into `01_Datasets\processed`. The dashboard uses the error fact table (so slicers work together). The CSV outputs are used for the report and to **cross-check dashboard totals**.

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